Temporal Network Analysis of School Interactions: A Study of Social Dynamics Across Grades

COMP0123 Complex Networks and Online Social Networks

Marcelina Marjankowska

Abstract

In this study, we analyze the temporal interaction network of primary school students to investigate the formation of social groups and the distinctions between lower and higher-grade students. Using high-resolution data collected via wearable sensors, which recorded face-to-face proximity events over two school days, we explore interaction patterns during unmoderated break periods to ensure the observed contacts reflect spontaneous social behavior. Our research addresses the challenge of interpreting proximity-based interaction data as indicators of meaningful social ties and investigates community structures formed within and across grades. We compare measures of temporal efficiency, average path lengths, and community modularity to assess the cohesion and mixing of interaction groups. Results indicate that younger students form less modular and more gender-mixed communities, while older students exhibit more defined and homogeneous social clusters. These findings align with broader sociological observations and highlight the utility of temporal network analysis for understanding children's social dynamics. This work contributes to the growing understanding of social network formation in educational environments and the methodological challenges posed by temporal data.

1 Introduction

We will analyze a temporal network of interactions among children in a primary school in Lyon, France. The network records face-to-face interactions between children. This dataset, developed by the SocioPatterns collaboration [Pro], was utilized in [Ste+11]. Its primary aim was to collect high-resolution data for applications, including the analysis of infectious disease spread.

In this project, I want to explore whether there is a distinction between interactions and community structures formed by higher-grade student and lower-grade ones. In particular, I aim to investigate whether distinct social groups form within these grades, how large these groups tend to be, and whether they exhibit any notable differences in terms of size or cohesion. By analyzing interaction patterns and the overall distribution of connections, I hope to uncover whether the underlying social dynamics vary significantly across grade levels.

To date, most analyses of the Lyon primary school interaction network have focused on modeling infectious disease spread and evaluating potential preventive measures that schools could implement [GBC14; Ste+11]. Here, I adopt a different approach by leveraging the available dataset to take a more sociological perspective. Many studies that examine group formation in schools rely on observations, self-reported data or friendship surveys, which are prone to bias and limited in scale [Cat+10; Lad83; MFB15]. By contrast, using an interaction network provides a more objective view and allows for the observation of a larger group of children.

Another reason why I believe it is worth investigating these questions is that temporal networks are highly prevalent in real life. From the World Wide Web to social interactions, communication networks, and transportation systems, most meaningful networks exhibit temporal dynamics. However, common methods for analyzing these networks often rely on time-aggregated views or discrete snapshots, which can result in significant information loss. It is crucial to leverage the temporal aspects such as time ordering, frequency, and duration of interactions. This is particularly important for dense networks, such as children's interaction networks, where a substantial amount of information may be lost in simplified representations. Therefore, it is valuable to evaluate the current capabilities and limitations of temporal network analysis tools.

As far as I understand, this research question is novel in relation to the available data. There are studies analyzing formation of social groups in children, but they often rely only on data based on observation of small

groups.

Understanding the social dynamics in schools can offer valuable insights into children's psychological and social development. This research could have practical applications, such as tailoring school programs to foster inclusivity and belonging. Insights into social dynamics can inform how schools structure group activities, seating arrangements, and collaborative projects to encourage healthy interactions.

In addition, differences in the types and strengths of connections between grades can inform broader theories about the formation and evolution of social networks. Since social group dynamics play a crucial role in children's sense of belonging and self-esteem, this research can also be relevant to psychologists interested in studying the relationship between social structures and psychological well-being.

I begin by reviewing the relevant literature. In Section 3, I introduce the network in detail and outline the methodology used to address the research question. Section 4 presents the numerical experiments. Finally, in Section 5, I conclude by discussing the results in greater depth and proposing directions for future work.

2 Literature survey

The dataset used in this analysis was introduced by Isella et al. (2009) [Ste+11], which examined the time-resolved face-to-face interactions of children and teachers in a French school environment. On average, each child had 323 contacts per day with 47 other children, with most interactions occurring within the same class. These findings challenge the homogeneous mixing assumption in disease models, suggesting more complex interaction structures. The dataset has been used to model the spread of infectious diseases [GBC14], but the sociological aspects of the interactions were not explored. This dataset provides a foundation for analyzing community dynamics, particularly the role of grade-level interactions, something previous studies have not explored. Existing work focuses on disease modeling, missing deeper insights into social structures that I aim to investigate.

Crowd interaction studies, particularly in disease spread, information diffusion, and connectivity, have been explored in temporal networks. A study by Isella et al. [Ise+11] on wearable sensor data from a conference and a museum highlighted differences between environments—conference networks were dense and small-world-like, while museum networks were fragmented. Temporal dynamics were shown to be crucial for understanding spreading

processes, as static networks can mislead by failing to capture causality. This inspired my analysis of contact durations in Section 4.1. The study emphasizes the importance of temporal factors, which align with my focus on the dynamics of interactions in a school setting. While the study examines spreading, it does not explore community dynamics, which is central to my analysis.

The reliability of proximity-based interactions in indicating friendship or social ties has been explored in various studies. For example, [MFB15] compares face-to-face contact patterns measured via wearable sensors, contact diaries, and friendship surveys in a French high school. The study highlights the strengths and limitations of these methods, showing that while wearable sensors are effective at capturing short interactions, longer interactions are more likely to represent meaningful friendships. Moreover, the study demonstrates that sensor-based networks, with their higher sampling rates, better reflect the temporal dynamics of interactions compared to contact diaries and surveys. However, the study also notes the limitations of these networks in interpreting the nature of interactions and defining what constitutes a friendship.

3 Methodology

We will use the Primary school temporal network data collected by the SocioPatterns collaboration [Pro]. The data set records interactions between primary school children recorded over two days. It has 242 nodes (of 232 children and 10 teachers) and interactions are recorded every 20 seconds. Each node has metadata about their gender and grade. There are five grades, each including two classes A and B, resulting in kids being assigned into 10 different classes.

The dataset was gathered through wearable sensors that detect close-range face-to-face proximity (1-1.5 meters) of participants who wear the sensors on their chests. The sensors are calibrated to detect face-to-face proximity between two individuals within a 20-second interval with a probability greater than 99%.This was designed so that when looking at the 20 second resolution, we don't encounter false negatives.

The school day commences from 8:30 to 17:00 with a lunch break between 12:00 and 14:00 during which some of the children leave school, and the data is not being collected. There are also two 30 minute breaks happening around 10:00 and 16:00. Because of facility restrictions different classes have a different schedules and I have analyzed the data to establish the

exact times for each of the four relevant grades. The details of that can be found in Appendix A. In the experiments I will analyze only the data from the first break during day 1, later referred to as 'the break'. I believe this is a good choice since all the children are present in the school, 30 minutes is enough time to reveal social patterns and most importantly, during breaks children are free to interact as they wish and they are not moderated by the teachers or sat in a classroom. If the whole day was analyzed, we would have the bias resulting from different teaching styles, lack of data resulting from children going home to lunch and interactions during lessons being forced rather than self-formed. I think that restricting to this shorter period results then in more interpretable results.

To investigate the proposed research question, I will focus on examining the interactions and community structures within the lowest and highest grades, specifically in classes 1A, 1B, 5A, and 5B. This selection ensures an age gap significant enough to highlight potential differences in social dynamics. Furthermore, the inclusion of two classes per grade level enables a comparative analysis, helping to verify that any observed patterns are not unique to a single class but are consistent across the age group.

Network Construction and Interaction Aggregation I will analyze four distinct networks, each representing one grade (1A, 1B, 5A, and 5B). Each network will include only within-grade interactions and will focus specifically on the break period of day 1. To account for the continuity of interactions, I will aggregate consecutive interactions occurring across successive timestamps. Interactions spanning multiple 20-second intervals will be treated as a single interaction, with their durations summed. These aggregated interactions will form the edges of the network, with weights corresponding to the total interaction duration.

Temporal Distance To take into account the temporal property of the network, we will consider the *temporal distance* of nodes, as defined in [Tan+09]. It is a distance measure that from a fixed starting time calculates minimal amount of timestamps to propagate an information from one node i to another j, denotes as d_{ij}. Parameter h specifies how many 'hops' can be taken within the same timestamp and in the following experiments I have used the value of $h = 3$. In particular, we can also define the *temporal efficiency* E between nodes i and j as the reciprocal of the temporal distance

$$E_{ij} = \frac{1}{d_{ij}}.$$

We can also calculate temporal measures for the full network such as *Average Temporal Path Duration*

$$L = \frac{1}{N(N-1)} \sum_{ij} d_{ij}$$

and the *Average Temporal Efficiency*

$$E_{\text{glob}} = \frac{1}{N(N-1)} \sum_{ij} E_{ij}$$

Temporal distance and temporal efficiency are key metrics that allow us to explore the dynamics of interaction over time. Temporal distance helps measure how quickly information (or interaction) propagates across the network, providing insights into how closely-knit different groups of students are. Temporal efficiency, on the other hand, offers a measure of how efficiently the network operates in terms of interaction spread. These metrics provide a way to quantify social cohesion and community strength.

Community Detection is a problem of finding densely connected communities of nodes in a network. Popular methods include the Girvan-Newman Algorithm and Clique Percolation Method. Here, we found that the Louvian method was the best fit. Introduced by Blandel et.a l. in 2008 [Blo+08] it is a greedy algorithm intending on maximizing the modularity. Modularity is a well-established metric in network science that quantifies the strength of division of a network into communities. Maximizing modularity helps identify subgroups of students who interact more frequently within their own community than with others.

Code for all of the performed experiments is available online at `https://github.com/marjankowskam/COMP01234_coursework`.

4 Results

In this section, we present experiments designed to explore similarities and differences between different grades. We begin by analyzing the aggregated network of interactions during school break. Table 1 summarizes basic properties of these networks, including node counts, average degrees, edge density, and average edge weights. Most important to note is that these aggregated networks are dense, with edge densities around 50%.

grade	1A	1B	5A	5B
nodes	21	25	20	23
average degree	12.5	17.44	9.1	11.4
edge density	52%	73%	39%	47%
average weight	78.5	107	68	93.5

Table 1: Basic metrics of the aggregated weighted networks during the break period.

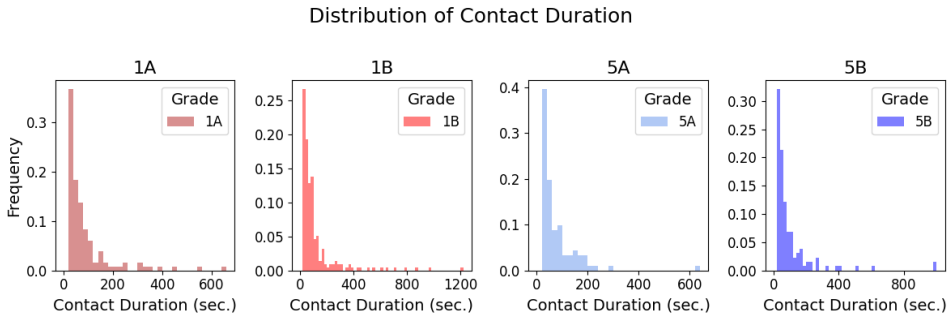


Figure 1: Distribution of contact durations in the four grades.

4.1 Contact Duration Analysis

Each contact can be characterized by its duration - we can aggregate consecutive interactions if they occur between the same two people. Plot 1 shows the distribution of the interaction durations. Most of the contacts are shortly-sustained lasting less than a minute. In particular, grade 5A has almost all interactions below 300 seconds. When plotted on a log-log scale, see Figure 2, we can observe the power law with an addition of a fatter tail. (Note here that the R^2 value is high enough to indicate that the linear fit is good.) This shape is a common observation in social contact networks [Ise+11; Cat+10].

We also consider the correlation between the number of distinct interactions and the average time spent interacting. This relationship is illustrated in Figure 3, where each point represents a child. Based on previous findings [Ise+11], we expect that individuals with a greater number of interactions would tend to have shorter average interaction durations, as they distribute their limited interaction time across more contacts. Interestingly, while a few 5th graders demonstrate longer average interaction durations with fewer number of interactions, this pattern is less apparent for other children, par-

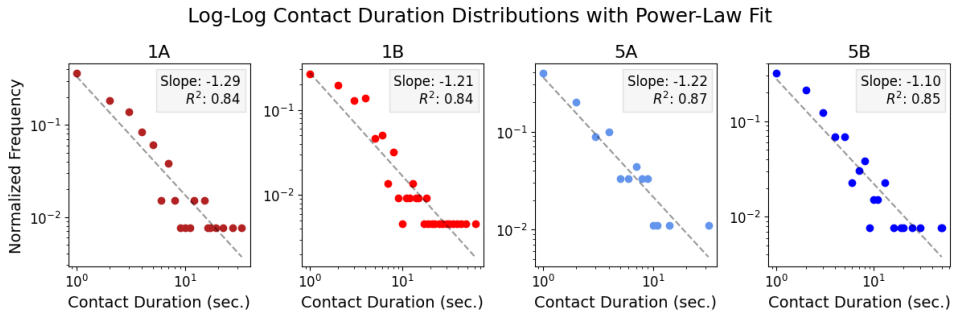


Figure 2: Distribution of the contact durations in the four grades. We can see the points following a linear trend with the R^2 metric indicating a good fit. This confirms that this distribution follows the power law.

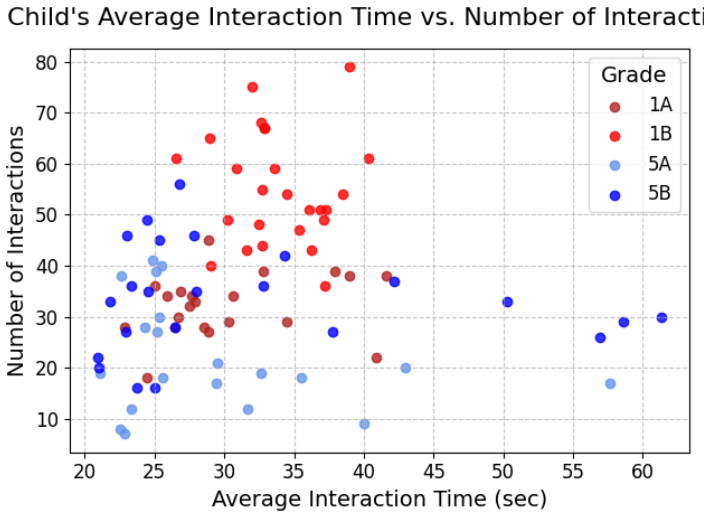


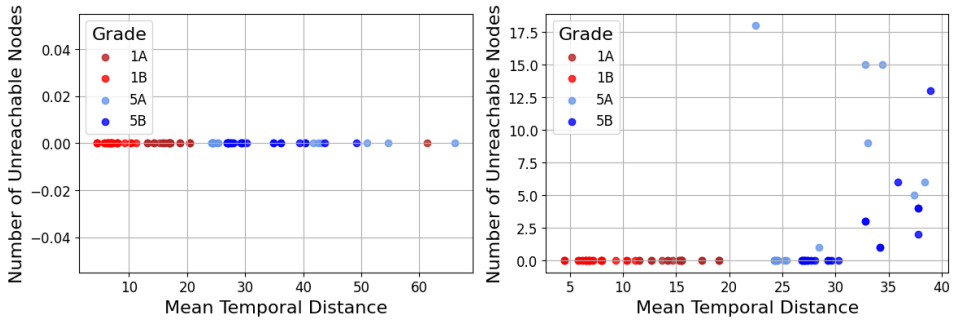
Figure 3: The scatterplot presents the average interaction time and the number of interactions for each child, color-coded by the grade assignment.

ticularly those with shorter average interactions. This suggests that some children might simply engage less overall.

A notable distinction emerges between younger and older students: younger children tend to have on average shorter but more frequent interactions, while older students display a wider range of interaction patterns. This difference likely reflects developmental or social changes in how contacts occur across grade levels.

4.2 Temporal Distances

We now analyze the mean temporal distances of nodes. It is worth mentioning that its value is proportional to the node's *Temporal Closeness Centrality* as defined in [Tan+10]. During the break time we run the temporal distance algorithm on the first 15 minutes and 30 minutes for each of the grades. So, for each node we end up having its temporal distance from all the other children in the given grade. Out of that we extract the average distance and the number of infinite distances (unreachable nodes). Results can be seen on plot 4. The difference between the first and fifth grades is clear showing that we end up with children in lower grades being within a smaller temporal distance of each other. At the same time, more of the higher graders are not reachable in the first 15 minutes suggesting social cliques and less mixing.



(a) first 30 minutes of the break (b) first 15 minutes of the break

Figure 4: Each point represents a node, with the x-value being the mean temporal distance to reachable nodes and the y-value being the count of unreachable nodes (infinite temporal distances). We are using the hyperparameter $h = 3$.

To further investigate these patterns, we consider the temporal efficiency

of these networks. To avoid relying on a fixed start time, we run the temporal distance algorithm with start times staggered every 2 minutes during the lunch break and calculate the average temporal efficiency of the network. The results are shown in Figure 5, where each point represents the network's temporal efficiency at a given moment. These results align with what we would expect based on previous experiments.

A clear difference emerges between the two age groups: the contact network of the lower-grade students consistently exhibits significantly lower efficiency. Additionally, the values are distributed over a wider range, indicating that the network's structure varies throughout the break period. This highlights how relying on a fixed start time could be misleading, as it would fail to capture the dynamic nature of the network.

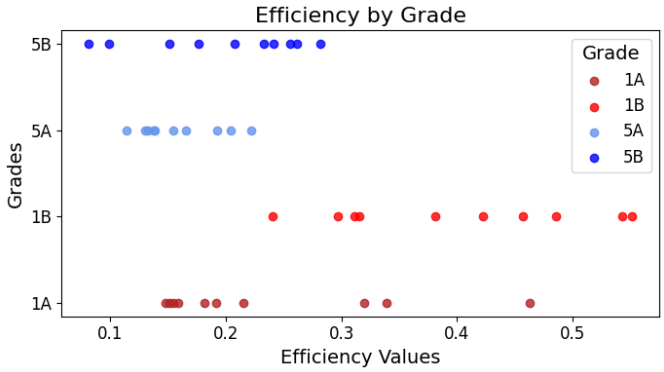


Figure 5: The network efficiency calculated as average of its nodes efficiencies. The value is calculated every 2 minutes during the break.

4.3 Community Detection

We now turn our attention to the formation of social groups within the grade groups under consideration. To investigate this, we apply the Louvain community detection method on weighted networks. These networks are constructed for each grade, with edge weights aggregated over the 30-minute break period. The edge weights, representing the total interaction time between pairs of children, provide a robust metric for detecting communities in these nearly complete graphs. This approach is designed to identify children who have frequent encounters or extended interactions while disregarding short, insignificant contacts.

Figure 6 illustrates the detected communities, with the resulting modu-

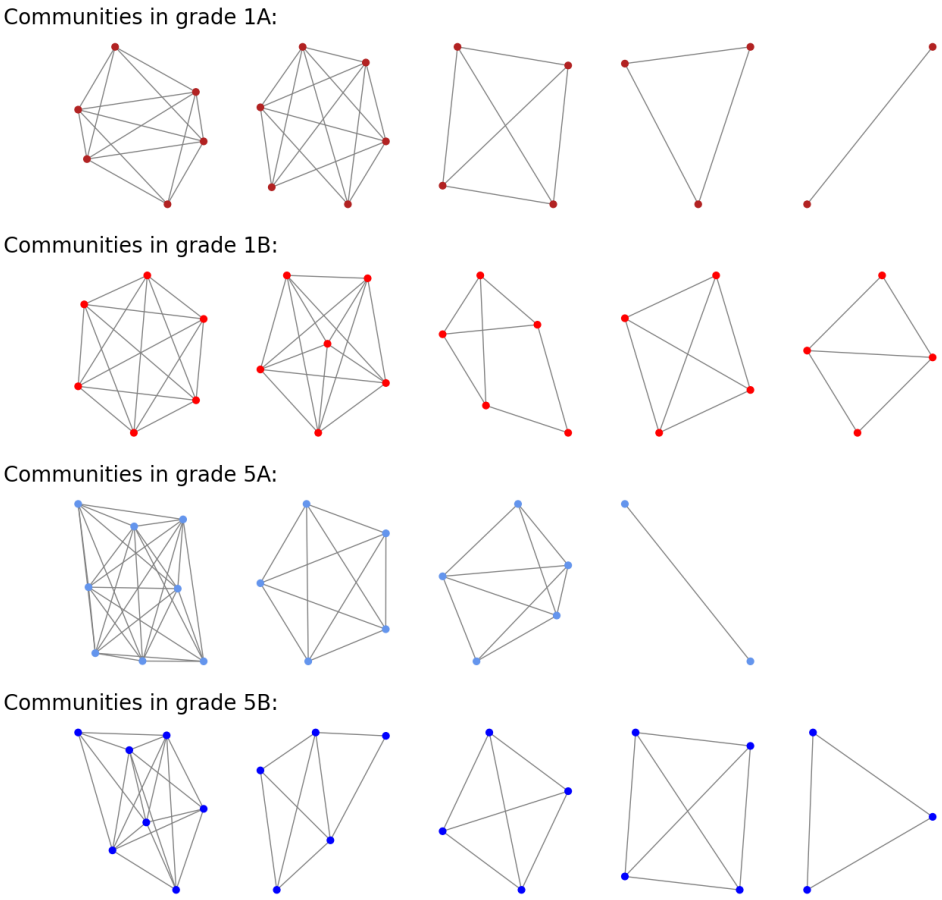


Figure 6: Communities detected within grades by the Louvian method with weights being the total contact duration during the break.

larity values as follows: 1A: 0.36, 1B: 0.35, 5A: 0.41, 5B: 0.45. An immediate observation is that lower-grade students form communities with lower modularity, suggesting that higher-grade students tend to organize into more distinct and well-defined social groups. This could indicate that as children grow older, their social interactions become more selective, leading to more strongly divided communities.

The Louvain algorithm detected between 4 and 5 communities in each grade group, with sizes ranging from 2 to 8 individuals. Many of these communities correspond to cliques, reflecting tightly connected subgroups within the networks. Interestingly, while the modularity values differ, the overall sizes and numbers of communities are not strikingly different across grades. This suggests that while the strength of division between groups increases with age, the basic structure of the networks remains somewhat consistent. Other methods of community detection were also considered and are described in Appendix B.

grade	1A	1B	5A	5B
day 1	0.43	0.48	0.65	0.76
day 2	0.39	0.5	0.56	0.78
break day 1	0.66	0.61	0.82	0.75

Table 2: Fraction of the total time of same gender interactions over all time of interactions.

4.4 Same versus Mixed-Gender Interaction

We begin by analyzing the proportion of total interaction time spent between same-gender students compared to that between students of different genders. When looking at Table 2 two notable patterns emerge. First, during school breaks, children tend to interact with more children of the same gender than they do over the course of the entire day. This suggests that lessons, likely moderated by teachers, encourage or facilitate different interactions. Second, a significant difference is evident between younger and older students, with younger students spending a larger proportion of their time in mixed-gender settings. I considered normalizing the above results, as detailed in Appendix C; however, the overall gender ratios across grades are very similar, and normalization would not provide additional insight in this context.

This tendency toward mixed-gender interaction among younger students is also reflected in the structure of their social groups. Communities identi-

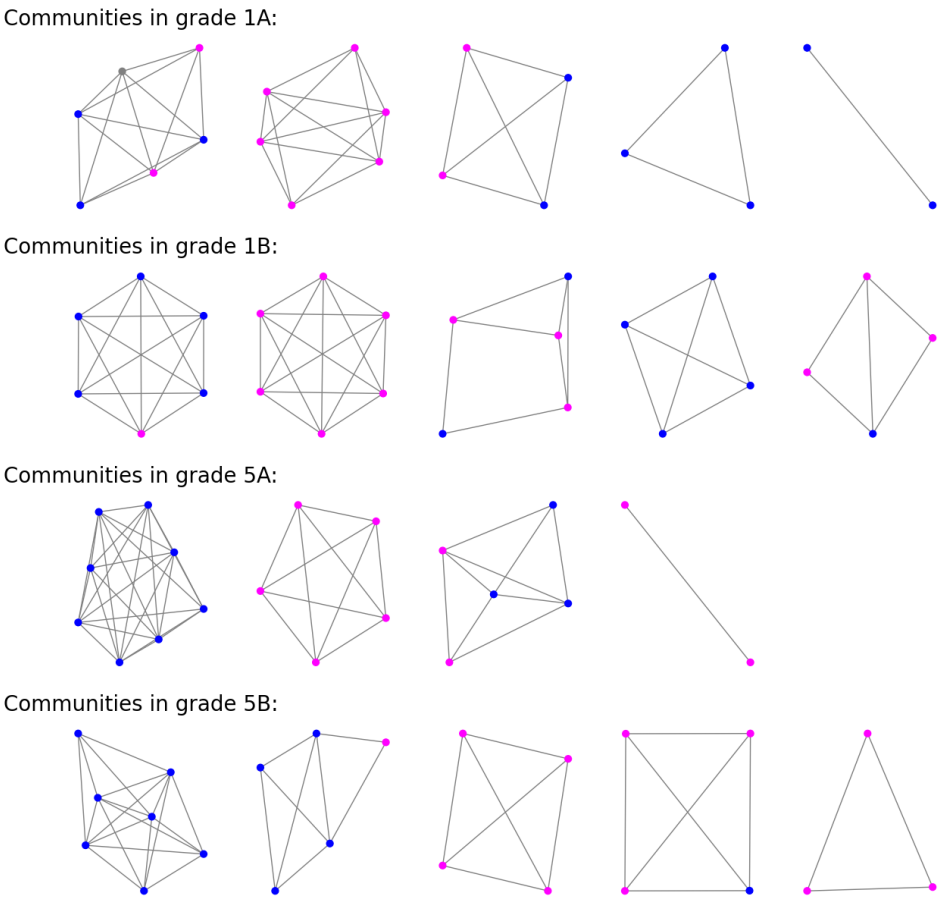


Figure 7: Communities detected within the grades by the Louvian method with weights being the total contact duration during the break. Pink nodes correspond to girls, blue to boys and the one gray node is not specified.

fied in the previous experiment were analyzed based on gender, as illustrated in Figure 7. Most communities are dominated by one gender, but there is slightly more mixing among younger students, particularly in Grade 1B, where 3 out of 5 clusters are "mixed." By contrast, in Grade 1A, 2 out of 5 clusters are mixed, while in Grades 5A and 5B, the figures are 1 out of 4 and 2 out of 5, respectively.

At first glance, this pattern might seem to contradict the data in Table 2, which shows higher levels of gender mixing in lower-grade students. However, the lower modularity values for these communities indicate that the boundaries between groups are less rigid, and there is still substantial interaction across groups.

An additional observation is the consistent presence of a large community predominantly composed of boys in each grade. This trend is particularly pronounced in the 5th-grade groups and aligns with findings from [Lad83], which indicate that boys tend to form significantly larger groups compared to girls.

5 Discussion

The results reveal several interesting patterns. Lower-grade students exhibit shorter temporal distances and higher reachability, suggesting a less hierarchical structure, which is further supported by the community detection analysis showing lower modularity. However, visual inspection reveals no significant differences in the size or number of formed communities. A clear pattern emerges with mono-gendered communities, especially among higher-grade students, with much more gender mixing observed in 1st-grade students compared to 5th-graders. Notably, this gender segregation is moderated by teachers, as shown by more balanced gender representation when considering the full day's data, rather than just the break period. These findings highlight the evolving structure of social networks as children age, with younger students forming more fluid and interconnected networks, while older students display more selective, stratified interactions. These results provide valuable insights for shaping educational strategies to foster more inclusive social environments. As seen in the experiments, looking at the data over the whole day can be misleading because of the moderation put in place by the teachers and varied activity types.

It is important to note that the appropriate approach to analyzing a temporal network depends on the context. For instance, in modeling infectious disease spread, the time-ordering of interactions is crucial, as properties like

causality and the lifespan of contacts play a significant role. In contrast, the problem of identifying social groups is more Markovian in nature, where the specific sequence of interactions is less important than their occurrence, co-occurrence, frequency, and duration. This is why I focused on these aspects in my analysis. Consider the notion of temporal distance as defined in this context—it is typically measured from a fixed starting point. Many temporal metrics similarly rely on a fixed start time. However, in some cases, using a single start time can introduce unnecessary variance in the results when that point is not meaningful. One way to address this issue is to compute the metric across multiple start times, as demonstrated in our node efficiency experiment.

Limitations One clear limitation of this study is the small sample size. The analysis focuses on a single 30-minute period and two grades for each age group, raising questions about the generalizability and statistical significance of the results. However, the observed patterns appear robust and offer valuable insights despite these constraints.

Limiting the analysis to within-grade interactions may overlook instances where children have friends in other grades and interact with them during breaks. However, this is unlikely to significantly affect the conclusions drawn about the within-grade dynamics.

Another limitation lies in the potential overestimation of interactions within the contact networks. The data may include false positives, where proximity is misinterpreted as meaningful interaction. For instance, children passing each other or sitting nearby might be recorded as having interacted, even when no substantive engagement occurred.

Further Work Future work could address these limitations by implementing strategies to reduce false positives. For example, interactions lasting 20 seconds or less could be excluded from the analysis to determine whether this adjustment alters the observed patterns. Additionally, expanding the dataset to include longer observation periods and more grades would help verify the consistency and significance of the findings.

Further work could also include analyzing more complex temporal metrics such as Temporal Rich Club [PBB22] and the Temporal Node Centrality metrics [Tan+09; Tan+10]. It would be also beneficial to analyze a larger amount of data to verify the significance of obtained results. I do believe that the results observed here are statistically significant; however, analyzing more time periods and observing more grades could solidify these beliefs.

Conclusion To summarize, in this report we have investigated the face-to-face interaction network of children in a Lyon primary school. We have observed differences in interactions within the highest and lowest grades in particular in the strength of formed communities. We observe younger children to mix more and have less defined social groups. The report considers the challenges inherent to temporal networks and presents strategies for analyzing these networks, which are frequently encountered in practical applications

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Appendix

A Breaks during the school day

Classes have different lesson schedules and so the exact time of their first 30-minute break differs. I have performed simple data analysis to infer the exact times for each of the grades 1A, 1B, 5A and 5B. To do this, I plot the number of interactions occurring, aggregated over 5 minutes for both within-grade interactions (blue line) as well as the cross-grade ones (red line) during the first day. The results can be seen in figure 8.

We can observe peaks in intensity indicating the two breaks and the lunch break. In addition, the peaks in cross-grade interactions (red line) indicate that the children are in common areas such as playground, gym, corridors or hall meaning that they are outside of their lesson period.

By inspecting the plots, I have arrived with the following times for the first break during day one 1A: 10:40:00, 1B: 10:10:00, 5A: 09:45:00, 5B: 09:45:00. The 30 minute intervals starting at these times, are then further used in the experiments.

B Other Community Detection methods

Because of the density of resulting networks, many of the community detection algorithms don't produce meaningful results. After much experimentation, I have found two noteworthy alternatives.

Girvan-Newman Alternative to the Louvian method used in this work, would be the Girvan-Newman method. Resulting communities can be seen in figure 9. I have decided to not use it as the resulting communities resulted in lower modularity suggesting that the method performed worse.

Cosine Similarity Inspired by [Ise+11] I have also considered an alternative to using the total interaction time as the edge weight. Instead, one can also use the cosine similarity. The cosine similiarity of two node is defined as

$$\mathrm{sim}_{ij} = \frac{\sum_{k \in \mathcal{V}} w_{ik} w_{jk}}{\sqrt{\sum_k w_{ik}^2 \sum_k w_{jk}^2}},$$

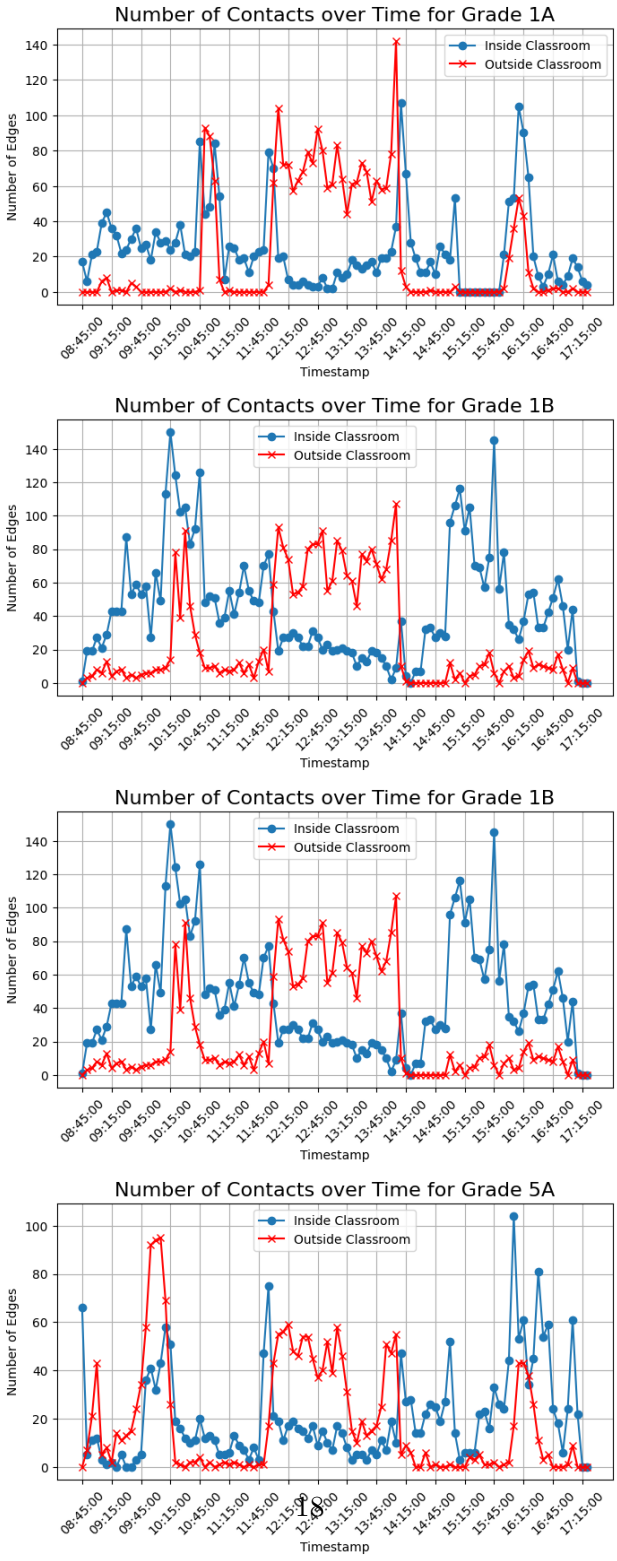

Number of Contacts over Time for Grade 1A
Inside Classroom
Outside Classroom
Number of Edges
Timestamp
Number of Contacts over Time for Grade 1B
Inside Classroom
Outside Classroom
Number of Edges
Timestamp
Number of Contacts over Time for Grade 1B
Inside Classroom
Outside Classroom
Number of Edges
Timestamp
Number of Contacts over Time for Grade 5A
Inside Classroom
Outside Classroom
Number of Edges
Timestamp

Figure 8

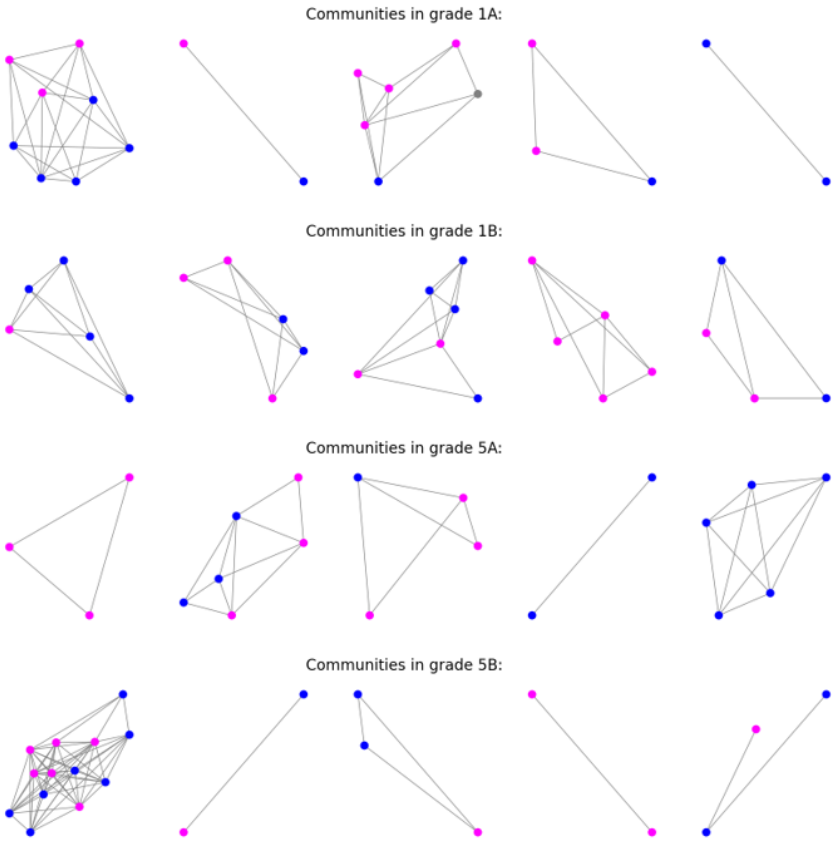


Figure 9: Communities detected within the grades by the Girvan-Newman method with weights being the total duration of contact.

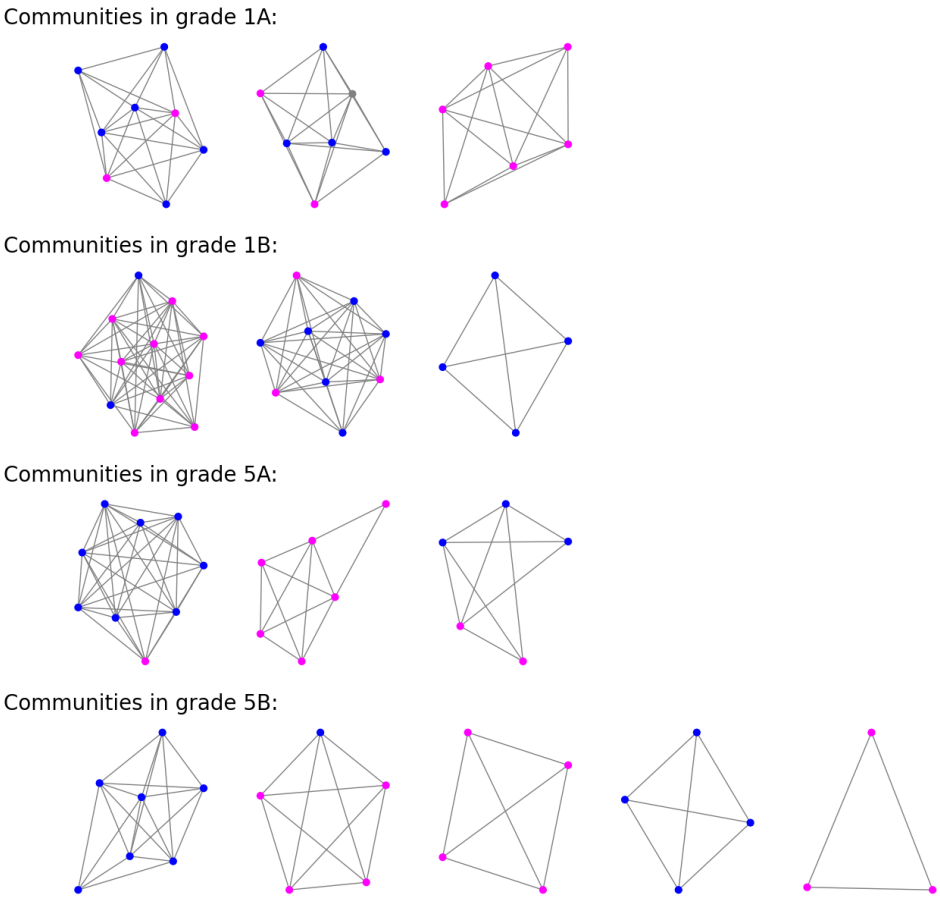


Figure 10: Communities detected within the grades by the Louvian method with weights being the cosine similarity between nodes.

where w_{ij} is the weight of the edge (i.e. total of contact duration) and $\mathcal{V}$ is the set of common neighbors. When these are used as weights, the Louvian method produces communities as shown in Figure 10.

C Normalization in Same vs. Mixed-Gender Interactions

An important consideration when analyzing same-gender interactions is whether the observed number of interactions should be normalized by the expected number of same-gender interactions, as this value is influenced by the gender ratio within each grade. From a mathematical perspective, let n represent the number of female students and m the number of male students. The expected fraction of same-gender interactions, assuming random interactions, is given by the formula:

$$\mathbb{E}[\text{fraction of same-gender interactions}] = 1 - \frac{n \cdot m}{(n+m)(n+m-1)}.$$

This follows from the fact that there are $(n+m)(n+m-1)/2$ possible edges in the interaction network, of which $n \cdot m$ represent edges between nodes of different genders. In a network where interactions are random, the expected fraction of same-gender interactions would follow this formula.

For the grades under consideration, the expected fraction of same-gender interactions is as follows:

- Grade 1A: 0.481
- Grade 1B: 0.480
- Grade 5A: 0.476
- Grade 5B: 0.478

As these values are very similar across the four grades, applying normalization would not significantly change the interpretation of the results. Then, for the sake of clarity, I chose not to include normalized results in the main analysis.